InsightAnalytics Group

PREDICTIVE INTELLIGENCE REPORT

EXPLAINABLE AI DIVISION

CONFIDENTIAL

EXPLAINABLE AI • PREDICTIVE MODELLING

Demand & Valuation Driver Analysis

A feature-level interpretability study using Partial Dependence Plots to identify the key variables driving bike-sharing demand and residential property valuations across two independent client engagements.

CLIENT A

Capital Bikeshare

Demand forecasting & driver analysis

CLIENT B

King County Real Estate

Property valuation driver analysis

NAVIGATION

Table of Contents

SECTIONS 01 & 02

Executive Summary & Methodology

Insight Analytics Group was engaged by two clients to produce a data-driven, interpretable analysis of the variables most strongly associated with their respective business outcomes: daily bike-sharing demand for Capital Bikeshare, and residential property sale prices for King County Real Estate. This report delivers those findings through **Partial Dependence Plots (PDP)**, a model-agnostic interpretability technique that quantifies the isolated marginal effect of each feature on the model's predictions.

KEY FINDING — BIKESHARE

Time trend and temperature are the primary demand drivers. Humidity suppresses demand above a 50% threshold. Windspeed has a consistent but secondary negative effect. Temperature dominates the joint humidity-temperature interaction.

KEY FINDING — REAL ESTATE

Living area is the strongest valuation driver with a wide monotonic effect. Bathrooms carry the largest absolute price range. Floors contribute positively but modestly. Bedroom count shows anomalous behaviour linked to data scarcity.

ANALYTICAL APPROACH

Random Forest models were fitted to each dataset. PDPs were computed over a 50-point grid per feature, marginalising over the full observed distribution of all other variables to extract globally-averaged feature effects.

Methodology

All predictive models were built using **Random Forest**, an ensemble learning method that aggregates predictions from hundreds of decision trees to produce robust, non-linear estimates. Random Forests were selected for their strong predictive performance and full compatibility with model-agnostic interpretability methods.

Interpretability was delivered through **Partial Dependence Plots**. For each feature of interest, its values are replaced by a regular grid of 50 evenly spaced points. At each grid point, the model predicts on all training observations with that substituted value, and the average prediction is recorded. This process isolates the feature's marginal contribution by averaging out the influence of all correlated variables.

> **Rug marks** displayed along the axes of each plot show the actual distribution of training observations, allowing the reader to distinguish regions of high data support from areas where the model is extrapolating. Predictions in data-sparse regions should be interpreted with appropriate caution.

Demand Driver Analysis — Individual Feature Effects

Capital Bikeshare provided daily operational records covering a two-year period from January 2011 to December 2012, comprising 731 observations enriched with meteorological measurements and calendar attributes. A Random Forest model trained on the complete dataset achieved an out-of-bag R^2 of **0.892**, confirming strong predictive performance and providing a reliable basis for the interpretability analysis.

DAILY RECORDS	MODEL ACCURACY	FEATURES ANALYSED	GRID RESOLUTION
731	**0.892**	**4**	**50 pts**
Full dataset	OOB R^2	Time, Temp, Humidity, Wind	Per feature

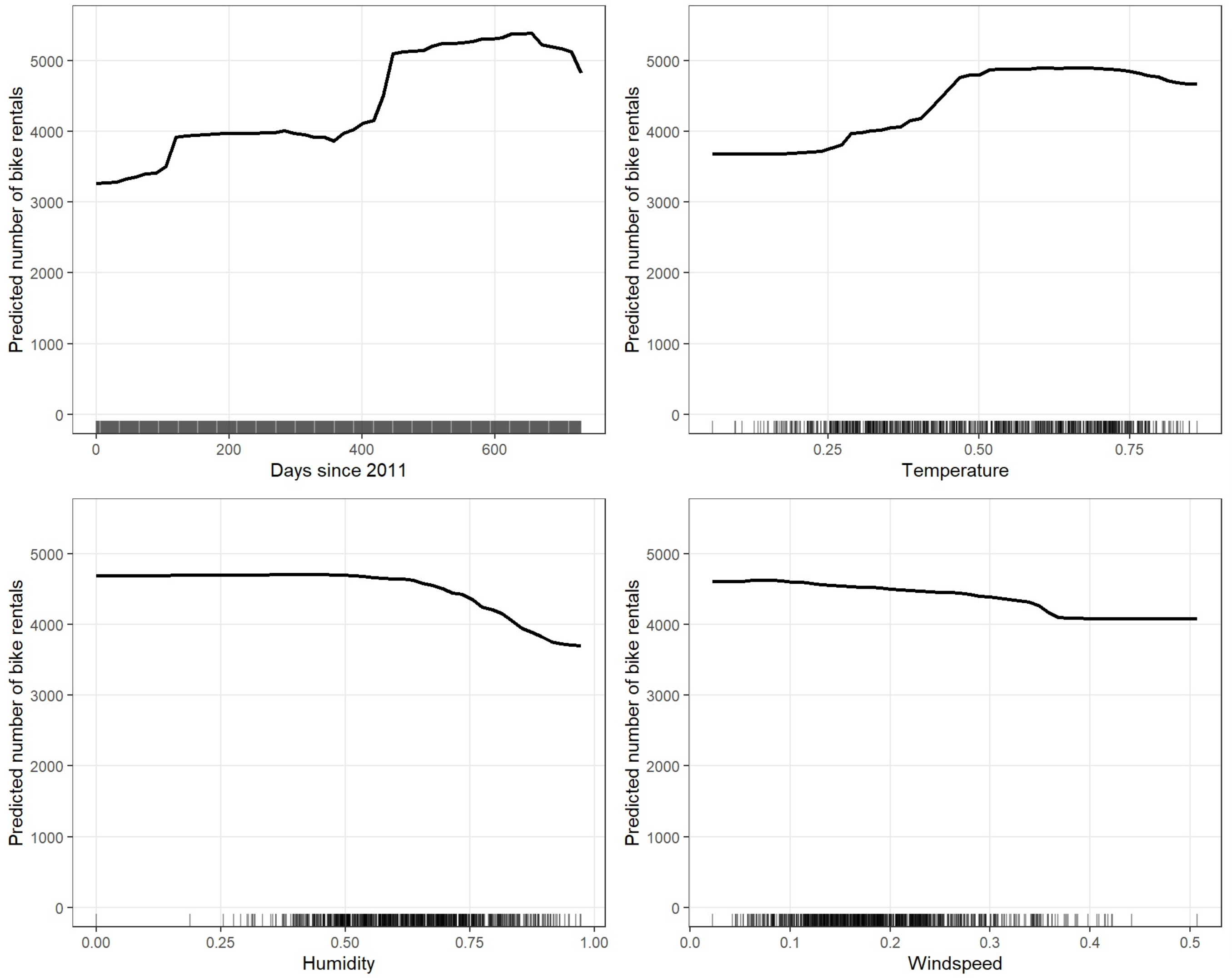

Figure 1. *Marginal effect of each demand driver on predicted daily rentals. Rug marks at the bottom of each panel indicate data density across the feature range.*

Findings

1 **Days since 2011 — Service adoption growth**

Predicted demand follows a stepped pattern: approximately 3,200 rentals in early 2011, stabilising around 4,000 through mid-year, then jumping sharply to 5,000–5,300 in 2012. This stepwise increase reflects strong year-on-year adoption growth, with the model distinguishing the two years as distinct demand regimes.

2 **Temperature — Saturating positive effect**

Demand rises steeply from approximately 3,600 rentals at low temperatures to a plateau of 4,800–5,000 at moderate-to-high values (normalised range 0.50–0.75), levelling off beyond that point. Comfortable cycling temperatures drive strong uptake, but very high heat provides no additional benefit.

3 Humidity — Threshold-gated suppression

Demand remains stable at approximately 4,700 rentals up to 50% humidity, then declines progressively to approximately 3,700 at maximum observed values. Users tolerate mild humidity well but are deterred once conditions become associated with rain and discomfort.

4 Windspeed — Moderate uniform deterrent

Rentals decline smoothly from approximately 4,650 to 4,100 across the observed windspeed range, with no identifiable threshold. The total effect of approximately 550 rentals is the smallest of the four features analysed, though consistent and monotone.

Demand Driver Analysis — Joint Effect of Humidity & Temperature

To understand how humidity and temperature interact in shaping demand, a two-dimensional Partial Dependence analysis was conducted across a 50×50 evaluation grid spanning the full observed range of both variables simultaneously. This surface reveals whether their combined influence produces interaction effects beyond what the individual analyses capture.

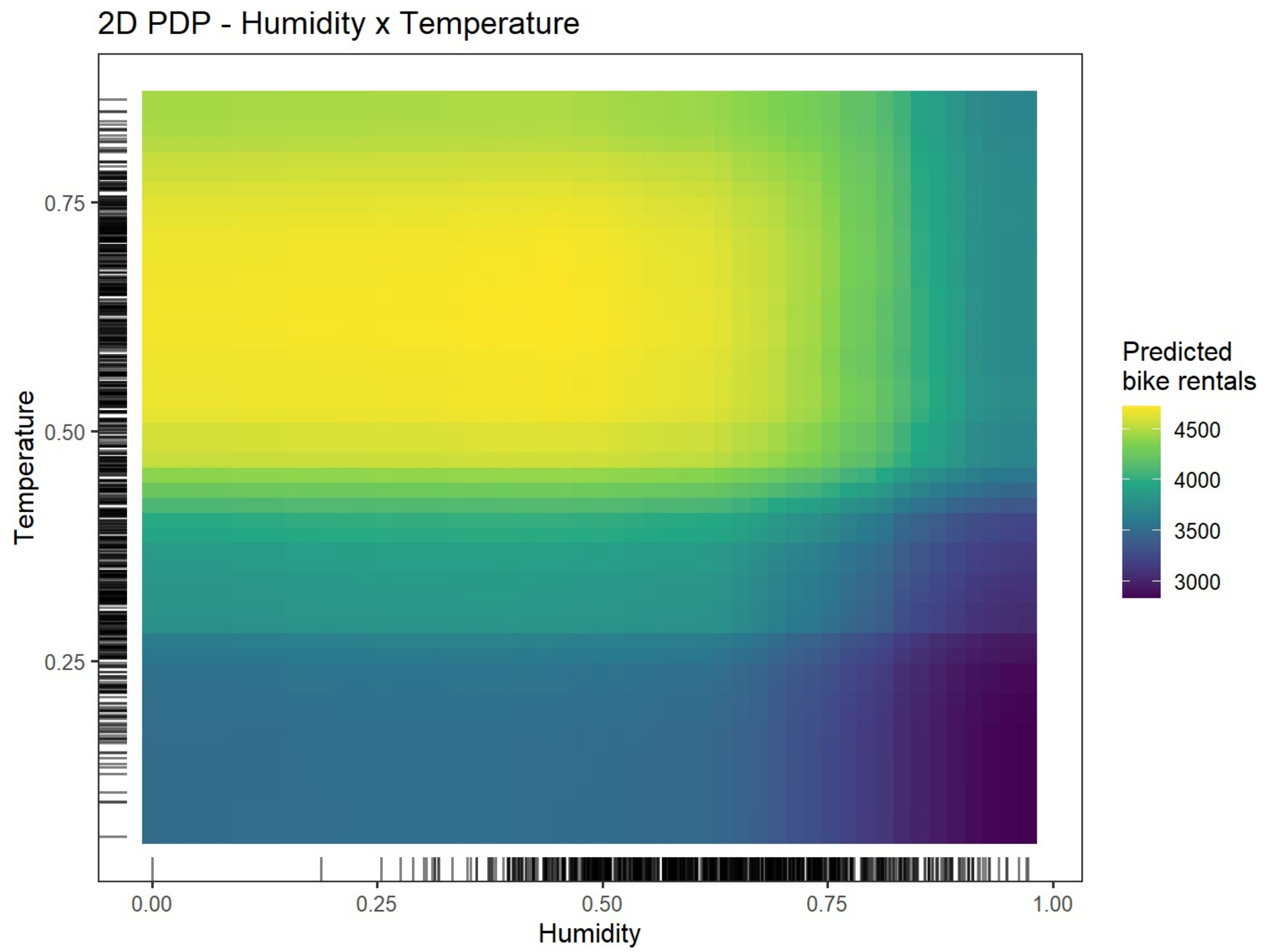

Figure 2. *Joint marginal effect of Humidity and Temperature on predicted daily rentals. Colour encodes predicted demand from low (dark purple, ~3,000) to high (yellow, ~4,500). Rug marks on both axes show the distribution of training observations.*

Interpretation

The colour gradient is **predominantly horizontal**, confirming that **temperature is the dominant driver** of the joint effect. High-demand zones (yellow bands) span the full width of the plot at temperatures above 0.50, irrespective of humidity. Low-demand zones (dark bands) similarly span the full humidity range at low temperatures — on a warm day, demand will be high regardless of humidity level.

Humidity's influence emerges only as a **modulating factor at high temperatures**: when temperature is high and humidity exceeds approximately 0.75, predicted demand shows a slight reduction relative to warm-and-dry conditions. This interaction is secondary and does not alter the fundamental temperature-driven structure of demand.

The rug marks reveal that real observations concentrate at medium-to-high humidity and moderate temperatures. The extreme corners of the surface are less supported by actual data and represent extrapolated conditions that should be interpreted with caution.

STRATEGIC RECOMMENDATION FOR CAPITAL BIKESHARE

Temperature is the single most actionable weather signal for anticipating demand. Operational decisions — fleet rebalancing, staffing allocation, maintenance scheduling — should be primarily indexed to temperature forecasts. Humidity warrants monitoring as a secondary risk factor, particularly above 50% on otherwise warm days.

The year-on-year step in demand also indicates that **capacity planning should account for secular adoption growth** alongside weather-driven seasonality, as the two effects operate on different timescales and must be modelled separately.

Valuation Driver Analysis — Individual Feature Effects

King County Real Estate provided a dataset of 21,613 residential property transactions. A representative random sample of **2,000 records** was used to train the predictive model — a standard practice for large datasets that preserves statistical representativeness while keeping computation tractable. The fitted Random Forest achieved an out-of-bag R^2 of **0.609**, a solid baseline given the complexity of real estate markets and the deliberately limited feature set provided.

TOTAL TRANSACTIONS	ANALYTICAL SAMPLE	MODEL ACCURACY	FEATURES ANALYSED
21,613	**2,000**	**0.609**	**4**
King County dataset	Random draw	OOB R^2	Beds, Baths, Sqft, Floors

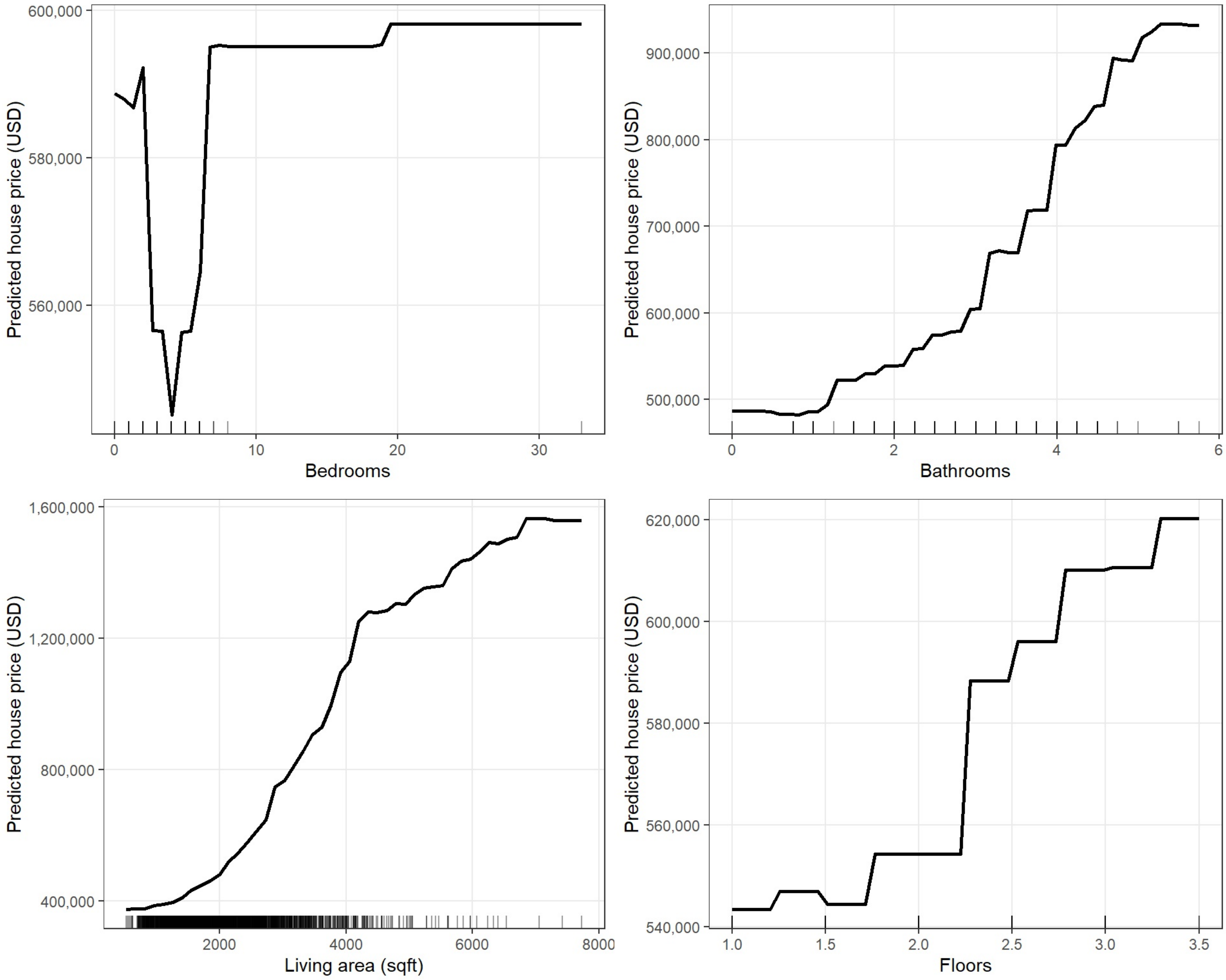

Figure 3. *Marginal effect of each property feature on predicted sale price. Rug marks indicate where the 2,000 training observations concentrate across each feature's range.*

Findings

1 Living Area (sqft) — Primary valuation driver

Predicted sale prices increase monotonically from approximately $370,000 for the smallest properties to $1,600,000 for the largest, representing the widest absolute range of any feature analysed. Growth is steepest between 1,000 and 4,000 sqft — where rug marks confirm the highest transaction concentration — and gradually levels off in the luxury segment.

2 Bathrooms — Largest absolute price impact

Predicted prices rise in a stepped pattern from approximately $490,000 for properties with one bathroom to $930,000 for those with five or six, a total range of approximately $440,000. Bathrooms function as a composite quality signal and represent the feature with the greatest absolute price differentiation across the portfolio.

3 Floors — Positive but moderate contribution

Predicted prices rise from approximately $543,000 for single-storey homes to $620,000 for 3.5-storey properties, with the sharpest increase between one and two floors. The total effect of approximately $77,000 is modest relative to living area and bathrooms, positioning floors as a contributing but not decisive valuation factor.

4 Bedrooms — Non-monotonic, data-limited signal

Bedroom analysis shows anomalous behaviour: prices start at approximately $590,000, fall sharply to a minimum around three to four bedrooms (~$548,000), then recover and stabilise near $595,000 beyond seven bedrooms. Rug marks reveal that observations concentrate almost entirely between one and six bedrooms, leaving insufficient data to learn a reliable effect at extreme counts.

STRATEGIC RECOMMENDATION FOR KING COUNTY REAL ESTATE

Living area and bathrooms are the two features with the strongest and most reliable influence on predicted sale prices. Valuation models, marketing strategies, and renovation investment decisions should prioritise these two dimensions above all others.

Bedroom count shows unreliable model behaviour beyond the most common range and should be cross-validated with richer data before being weighted heavily in automated valuation pipelines.

SECTION 05

Conclusions & Recommendations

Across both client engagements, Partial Dependence Plot analysis successfully translated complex Random Forest model behaviour into clear, actionable business insights. The technique effectively isolated the marginal contribution of individual features, revealing non-linear patterns, demand thresholds, and interaction effects that simpler linear approaches would fail to capture.

Summary of Findings

A

Capital Bikeshare

Demand is primarily governed by year-on-year adoption growth and temperature, with humidity acting as a threshold-gated suppressor above 50% and windspeed providing a consistent but moderate deterrent. In the joint analysis, temperature dominates the interaction with humidity, making it the most actionable operational signal for fleet and staffing decisions.

B

King County Real Estate

Property valuation is driven primarily by living area — monotonic, wide-ranging, and the strongest single predictor — and bathrooms, which show the largest absolute price impact. Floor count contributes positively but modestly. Bedroom count exhibits anomalous behaviour and should be used cautiously in automated systems pending richer data collection.

Methodological Considerations

Marginal Independence Assumption: PDPs assume that the feature of interest is independent of the remaining features during marginalisation. When features are correlated — as is the case for living area, bedrooms, and bathrooms in the real estate dataset — the grid evaluation may integrate over unrealistic combinations, potentially distorting the estimated effect. **Accumulated Local Effects (ALE) plots** provide a more statistically robust alternative and are recommended as a follow-up analysis for both datasets.

Next Steps

Insight Analytics Group recommends the following next steps for both clients: expanding the feature set to include additional operational and contextual variables; applying ALE plots to address feature correlation concerns; and integrating the identified key drivers — temperature for bikeshare demand, living area and bathrooms for property valuation — as primary inputs in real-time scoring and decision-support systems.